

1. Exercise 5.1.1. a
2. Exercise 5.1.7 , 5.4.1 and 5.4.3
3. Exercise 6.1.1. b
4. Consider the set of all strings of balanced parentheses of two types round and square. An example of where these strings come from is as follows. If we take expressions in C which use round parentheses for grouping and for arguments of function calls and use square brackets for array indexes and drop out everything but the parentheses we get all strings of balanced parentheses of these two types. For example $F(a[i] * (b[i] [j], c[g(x)]), d[i])$ becomes the balanced parenthesis string. Design a grammar for all and only the strings of round and square parentheses that are balanced.
5. Convert the following grammar into a normal form.

$$S \rightarrow CBC$$

$$C \rightarrow aC \mid \epsilon$$

$$B \rightarrow bBc \mid \epsilon$$

6. Check whether the following grammar is ambiguous or not

$$S \rightarrow aSb \mid SS$$

$$S \rightarrow \epsilon$$

7. Show that the given grammar is ambiguous. Also, find an equivalent unambiguous grammar.

$$S \rightarrow XYX$$

$$X \rightarrow aX \mid \epsilon$$

$$Y \rightarrow bY \mid \epsilon$$

8. Give pushdown automata that recognize the following languages. For each PDA, give both a state diagram and 6-tuple specification. Every context-free language has infinitely many (correct) PDAs, but you only need to give one.

$$A = \{ w \in \{0, 1\}^* \mid w \text{ contains at least three 1s} \}$$